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Project: 226052 DFCI-BIDMC New Cancer Hospital - 1 Joslin Place - Construction
1 Joslin Place
Boston, Massachusetts 02215

RFI #0-PB-33: Vault Ceiling Height

Revision	0	**Status**	Closed on 11/17/25
To	Rob Morsi (Suffolk Construction) *(Response Required)*	**From**	Nathan Crowley (Suffolk Construction) 65 Allerton St. Boston, Massachusetts 02119
Date Initiated	Nov 7, 2025	**Due Date**	Nov 14, 2025
Location		**Project Stage**	Bidding
Cost Impact		**Schedule Impact**	
Spec Section		**Cost Code**	
Drawing Number	S1.00, S1.00A, A1.00, A1.01	**Reference**	
Linked Drawings			
Received From		**Sub Job**	
Copies To			

ADDITIONAL FIELDS

Required Reviewers	**Urgency**

Activity

Question

Question from Nathan Crowley Suffolk Construction *on Friday, Nov 7, 2025 at 02:23 PM EST*

The slab edge plans provide a clear height of 15’ in the ground level Eversource vault, which meets the 15’ minimum height requirement per Eversource’s construction standard DTR 76.653, line item 7.2. The structural plans provide a clear height of 14’-3”.

1. Please confirm that the TOS above the vault is to be located at el 52’-0.
2. Please clarify whether the slab construction for the vault lid is to be 5-1/4” NW concrete on 3” deck or 4-1/2” NW concrete on 3” deck.
3. Please advise if any changes to the 3/S1.00A steel members or connection details are required based on the questions above.

Attachments
Vault Clear Height RFI.pdf

Official Response

Response from Henry Sniezek Suffolk Construction *on Monday, Nov 17, 2025 at 02:34 PM EST*

Per Payette 11/14 response: TOS above the vault shall be located at El. 52'-8.25"
Slab construction for the vault lid shall be 5.25" NWC on 3" deck.
There are not changes to the 3/S1.00A steel members or connection details based on responses above.
Changes to TOS and slab construction will be reflected in the 1/15 Addendum.

All Replies

Response from Henry Sniezek Suffolk Construction *on Monday, Nov 17, 2025 at 02:34 PM EST*

Per Payette 11/14 response: TOS above the vault shall be located at El. 52'-8.25"
Slab construction for the vault lid shall be 5.25" NWC on 3" deck.
There are not changes to the 3/S1.00A steel members or connection details based on responses above.
Changes to TOS and slab construction will be reflected in the 1/15 Addendum.

GENERAL NOTES:

G1. FOR COLUMN SIZES, SEE S2.00 THRU S2.05.
- INDICATES COLUMN STARTS.
- INDICATES CONTINUOUS COLUMN.
- INDICATES COLUMN STOPS.
- ETC, INDICATES CONNECTION TYPE OR STRUCTURAL USE. SEE DETAIL 1 / S2.05. FOR ADDITIONAL INFORMATION.

G2. FOR SLAB CONSTRUCTION, SEE PLAN NOTES.
- ETC, INDICATES STEP IN STRUCTURAL SLAB, SEE S0.04.
- ETC, INDICATES SPAN DIRECTION OF STEEL DECK.
- INDICATES SLAB OVERPOUR TO BE POURED MONOLITHICALLY WITH TYPICAL SLAB CONSTRUCTION (NO HORIZONTAL JOINT IN SLAB).
- INDICATES BONDED TOPPING SLAB. SEE DETAIL 4 / S0.30.
- INDICATES MECHANICAL OR MEDICAL UNIT.
- INDICATES MECHANICAL HOUSEKEEPING PAD.
- INDICATES OPENINGS WITHOUT POURSTOP. DECK OVER ENTIRE OPENING SHOWN. FOAM BLOCKOUT WILL BE PROVIDED WHERE REQUIRED AND DECK WILL NOT BE CUT. CONCRETE INFILL WILL BE PROVIDED OUTSIDE OF FOAM BLOCKOUT EXTENTS. SEE TYPICAL DETAILS FOR REBAR AND STEEL THAT MAY BE REQUIRED FOR FINAL OPENING EXTENTS.

G3.
- INDICATES GRAVITY MOMENT CONNECTION. SEE DETAIL 8 / S0.07.
- INDICATES MOMENT FRAME CONNECTION. SEE DETAIL 5 & 6 / S0.07.
- INDICATES OPENING IN WEB OF BEAM. SEE DETAIL 8 / S0.08.
- WT — INDICATES TORSION CONNECTION. SEE DETAIL 5 / S0.07.
- INDICATES WT BOLSTER ATOP LOW WF. SEE DETAIL 2 / S0.09.
- INDICATES MOMENT PLATE CONNECTION. SEE DETAIL 7 / S0.07.
- MP / TH — INDICATES THERMAL BREAK. PROVIDE REQUIRED THERMAL BREAK FOR FORCES NOTED. SEE ARCHITECTURAL DRAWINGS FOR THERMAL INSULATION MATERIAL. SEE DETAIL 1 / S0.12 FOR ADDITIONAL INFO.

G4. FRAMING ELEVATIONS CORRESPOND TO PARALLEL GRID LINE.
- **BF-1** — ETC, INDICATES BRACED FRAME. SEE ELEVATIONS ON DRAWINGS S2.10 THRU S2.11.
- ±20k ±20k — ETC, INDICATES LRFD HORIZONTAL THROUGH FORCE TO BE USED FOR BEAM/COLUMN CONNECTION DESIGN. SEE BRACED FRAME ELEVATIONS FOR VALUE WHERE NOT NOTED ON PLAN.

G5. SEE DRAWINGS S1.70 THRU S1.76 FOR ELEVATOR PART PLANS.

G6. ETC, INDICATES CONCRETE CURB. UNLESS NOTED OTHERWISE SEE REINFORCING 2 / S0.03.

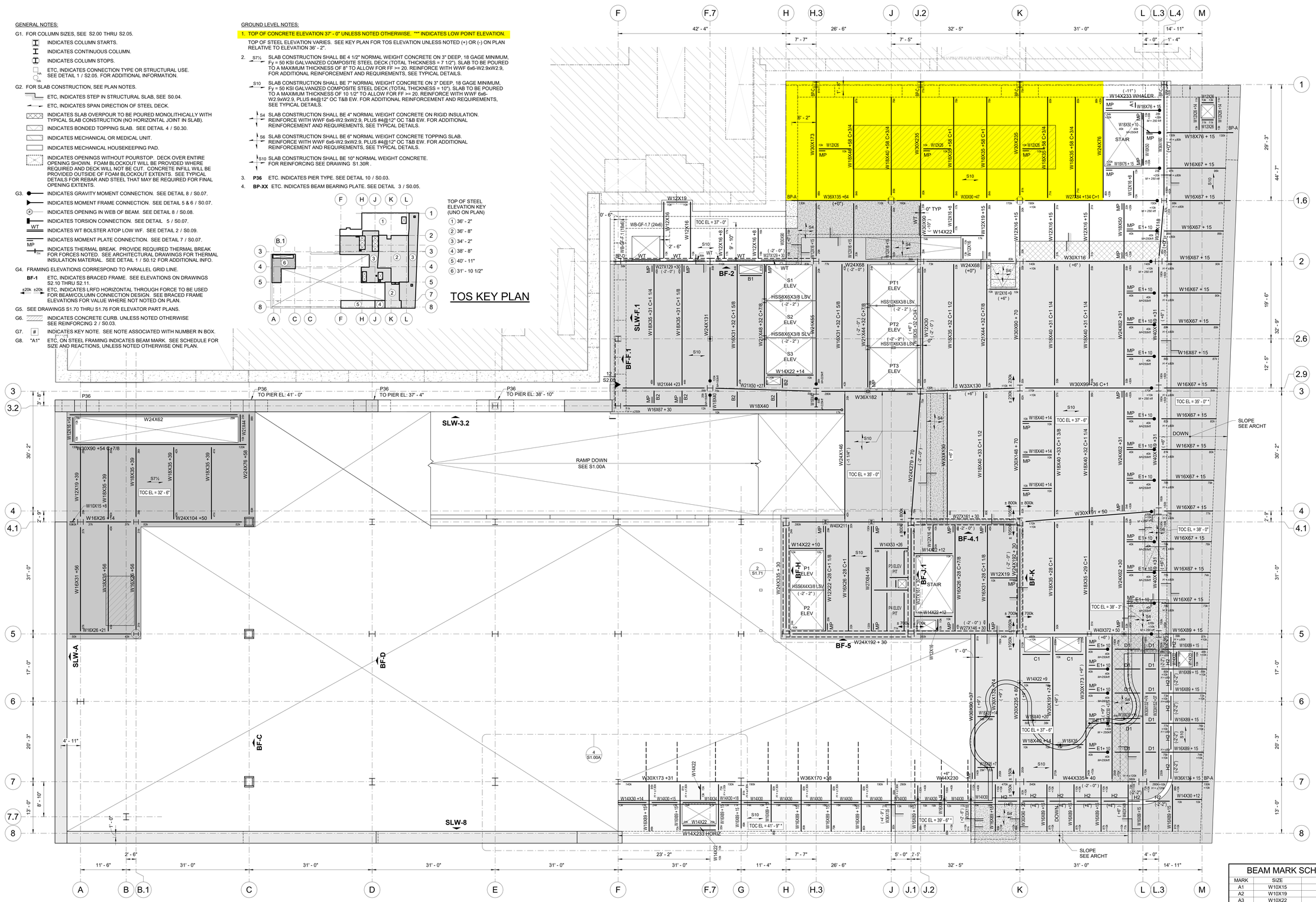
G7. # — INDICATES KEY NOTE. SEE NOTE ASSOCIATED WITH NUMBER IN BOX.

G8. "A1" — ETC, ON STEEL FRAMING INDICATES BEAM MARK. SEE SCHEDULE FOR SIZE AND REACTIONS, UNLESS NOTED OTHERWISE ONE PLAN.

GROUND LEVEL NOTES:

1. TOP OF CONCRETE ELEVATION 37' - 0" UNLESS NOTED OTHERWISE. *** INDICATES LOW POINT ELEVATION.
   TOP OF STEEL ELEVATION VARIES. SEE KEY PLAN FOR TOS ELEVATION UNLESS NOTED (+) OR (-) ON PLAN RELATIVE TO ELEVATION 36' - 2".
2. S7½ — SLAB CONSTRUCTION SHALL BE 4 1/2" NORMAL WEIGHT CONCRETE ON 3" DEEP, 18 GAGE MINIMUM, Fy = 50 KSI GALVANIZED COMPOSITE STEEL DECK (TOTAL THICKNESS = 7 1/2"). SLAB TO BE POURED TO A MAXIMUM THICKNESS OF 8" TO ALLOW FOR FF >= 20. REINFORCE WITH WWF 6x6-W2.9xW2.9, FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.
   S10 — SLAB CONSTRUCTION SHALL BE 7" NORMAL WEIGHT CONCRETE ON 3" DEEP, 18 GAGE MINIMUM, Fy = 50 KSI GALVANIZED COMPOSITE STEEL DECK (TOTAL THICKNESS = 10"). SLAB TO BE POURED TO A MAXIMUM THICKNESS OF 10 1/2" TO ALLOW FOR FF >= 20. REINFORCE WITH WWF 6x6-W2.9xW2.9, PLUS #4@12" OC T&B EW. FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.
   S4 — SLAB CONSTRUCTION SHALL BE 4" NORMAL WEIGHT CONCRETE ON RIGID INSULATION. REINFORCE WITH WWF 6x6-W2.9xW2.9, PLUS #4@12" OC T&B EW. FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.
   S6 — SLAB CONSTRUCTION SHALL BE 6" NORMAL WEIGHT CONCRETE TOPPING SLAB. REINFORCE WITH WWF 6x6-W2.9xW2.9, PLUS #4@12" OC T&B EW. FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.
   S10 — SLAB CONSTRUCTION SHALL BE 10" NORMAL WEIGHT CONCRETE. FOR REINFORCING SEE DRAWING S1.30R .
3. **P36** — ETC. INDICATES PIER TYPE. SEE DETAIL 10 / S0.03.
4. **BP-XX** — ETC. INDICATES BEAM BEARING PLATE. SEE DETAIL 3 / S0.05.

TOS KEY PLAN



BEAM MARK SCHEDULE (G-3)

MARK	SIZE	REACTIONS
A1	W10X15	20k SHEAR EE
A2	W10X19	20k SHEAR EE
A3	W10X22	20k SHEAR EE
B1	W12X19	20k SHEAR EE
B2	W12X16	10k SHEAR EE
B3	W12X22	20k SHEAR EE
B4	W12X26	20k SHEAR EE
B5	W12X30	20k SHEAR EE
C1	W14X22	10k SHEAR EE
D1	W16X50	10k SHEAR EE, ±80k AXIAL EE
E1	W18X50	40k SHEAR EE
H2	HSS12X10X3/8 LSH	Ry = 30k, Rx = 20k, & T = 60kft EE
L1	L4X4X3/8	10k SHEAR EE
T1	WT6X25	5k SHEAR & 15kft MOMENT

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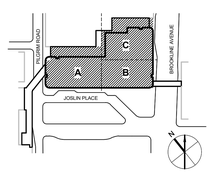
Issues

Number	Date	Description
00	10-30-25	FACADE AND INFRASTRUCTURE PACKAGE

Revisions

Number	Date	Description

NOT FOR CONSTRUCTION



Dana-Farber Cancer Institute
Beth Israel Lahey Health
Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL

1 JOSLIN PLACE, BOSTON MA 02115

GROUND LEVEL FRAMING PLAN

Drawing Scale: As indicated
Project Number: 17000.00
Date Issued: 10/30/2025

S1.00

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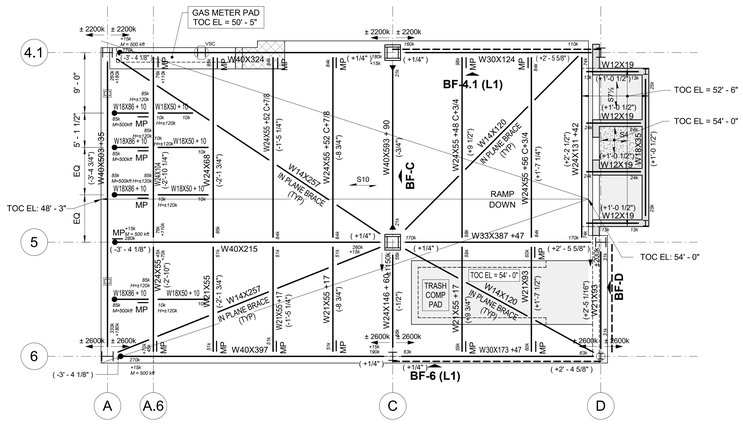
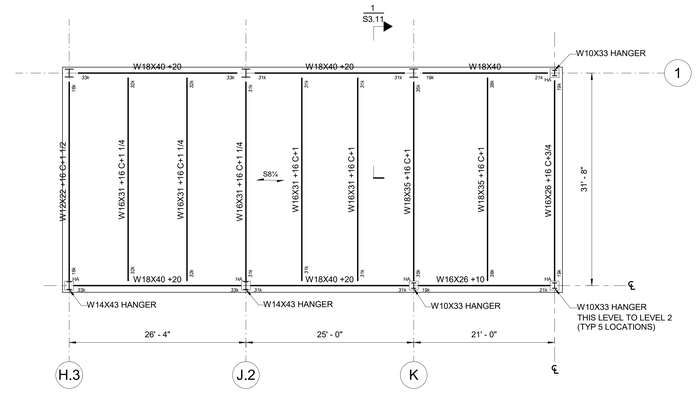


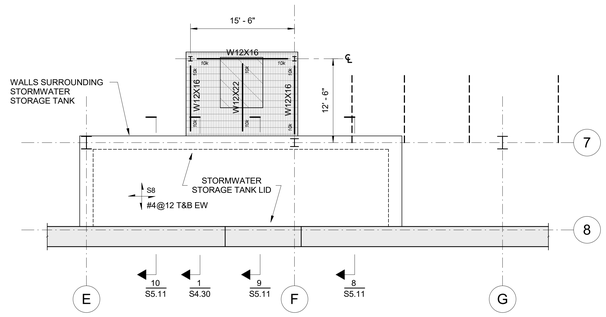
1 LOADING DOCK PART PLAN

SCALE: 1/8" = 1'-0"



3 ELEC VAULT LID PART PLAN

SCALE: 1/8" = 1'-0"



4 STORMWATER TANK CAP PART PLAN

SCALE: 1/8" = 1'-0"

GENERAL NOTES:

G1. FOR COLUMN SIZES, SEE S2.00 THRU S2.05.

- INDICATES COLUMN STARTS.
- INDICATES CONTINUOUS COLUMN.
- INDICATES COLUMN STOPS.
- ETC, INDICATES CONNECTION TYPE OR STRUCTURAL USE. SEE DETAIL 1 / S2.05. FOR ADDITIONAL INFORMATION.

G2. FOR SLAB CONSTRUCTION, SEE PLAN NOTES.

- ETC, INDICATES STEP IN STRUCTURAL SLAB, SEE S0.04.
- ETC, INDICATES SPAN DIRECTION OF STEEL DECK.
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- INDICATES OPENINGS WITHOUT POURSTOP. DECK OVER ENTIRE OPENING SHOWN. FOAM BLOCKOUT WILL BE PROVIDED WHERE REQUIRED AND DECK WILL NOT BE CUT. CONCRETE INFILL WILL BE PROVIDED OUTSIDE OF FOAM BLOCKOUT EXTENTS. SEE TYPICAL DETAILS FOR REBAR AND STEEL THAT MAY BE REQUIRED FOR FINAL OPENING EXTENTS.

G3.
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- INDICATES MOMENT FRAME CONNECTION. SEE DETAIL 5 & 6 / S0.07.
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G4. FRAMING ELEVATIONS CORRESPOND TO PARALLEL GRID LINE.

- **BF-1** ETC, INDICATES BRACED FRAME. SEE ELEVATIONS ON DRAWINGS S2.10 THRU S2.11.
- ±20k ±20k — ETC, INDICATES LRFD HORIZONTAL THROUGH FORCE TO BE USED FOR BEAM/COLUMN CONNECTION DESIGN. SEE BRACED FRAME ELEVATIONS FOR VALUE WHERE NOT NOTED ON PLAN.

G5. SEE DRAWINGS S1.70 THRU S1.76 FOR ELEVATOR PART PLANS.

G6. INDICATES CONCRETE CURB. UNLESS NOTED OTHERWISE SEE REINFORCING 2 / S0.03.

G7. # INDICATES KEY NOTE. SEE NOTE ASSOCIATED WITH NUMBER IN BOX.

G8. "A1" ETC, ON STEEL FRAMING INDICATES BEAM MARK. SEE SCHEDULE FOR SIZE AND REACTIONS, UNLESS NOTED OTHERWISE ONE PLAN.

LOADING DOCK AND DOCK LEVELER NOTES:

1. TOP OF CONCRETE ELEVATION VARIES, SEE PLAN.

   TOP OF STEEL ELEVATION 50' - 10" UNLESS NOTED (+) OR (-) ON PLAN.

2. S7½ — SLAB CONSTRUCTION SHALL BE 4 1/2" NORMAL WEIGHT CONCRETE ON 3" DEEP, 18 GAGE MINIMUM, Fy = 50 KSI GALVANIZED COMPOSITE STEEL DECK (TOTAL THICKNESS = 7 1/2"). SLAB TO BE POURED TO A MAXIMUM THICKNESS OF 8" TO ALLOW FOR FF >= 20. REINFORCE WITH WWF 6x6-W2.9xW2.9, FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.

   S10 — SLAB CONSTRUCTION SHALL BE 7" NORMAL WEIGHT CONCRETE ON 3" DEEP, 18 GAGE MINIMUM, Fy = 50 KSI GALVANIZED COMPOSITE STEEL DECK (TOTAL THICKNESS = 10"). SLAB TO BE POURED TO A MAXIMUM THICKNESS OF 10 1/2" TO ALLOW FOR FF >= 20. REINFORCE WITH WWF 6x6-W2.9xW2.9, PLUS #4@12" OC T&B EW. FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.

ELEC VAULT LID NOTES:

1. TOP OF CONCRETE ELEVATION 51' - 11 1/4" UNLESS NOTED OTHERWISE.

   TOP OF STEEL ELEVATION 51' - 3" UNLESS NOTED (+) OR (-) ON PLAN.

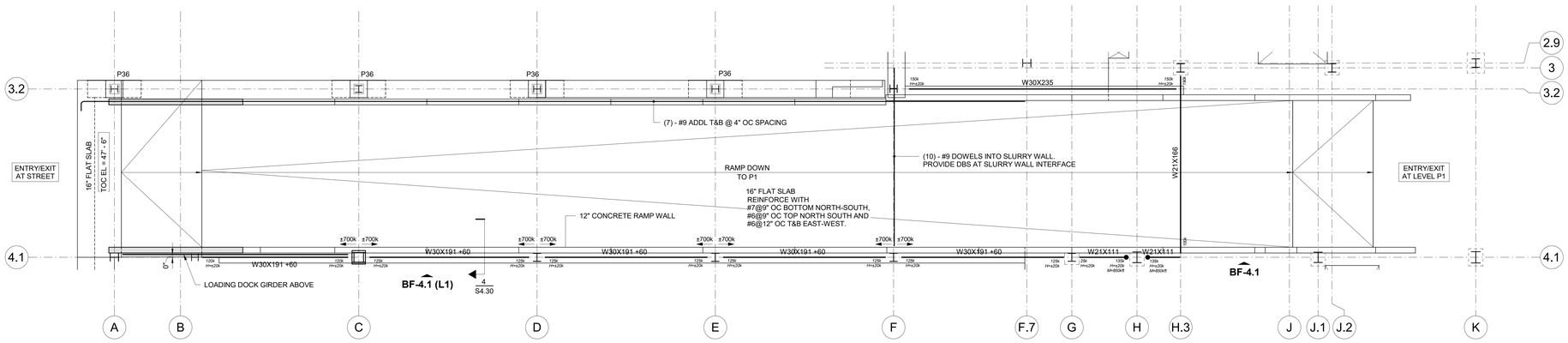
2. S8¼ — SLAB CONSTRUCTION SHALL BE 5 1/4" NORMAL WEIGHT CONCRETE ON 3" DEEP, 18 GAGE MINIMUM, Fy = 50 KSI GALVANIZED COMPOSITE STEEL DECK (TOTAL THICKNESS = 8 1/4"). SLAB TO BE POURED TO A MAXIMUM THICKNESS OF 8.75" TO ALLOW FOR FF >= 20. REINFORCE WITH WWF 6x6-W2.9xW2.9, FOR ADDITIONAL REINFORCEMENT AND REQUIREMENTS, SEE TYPICAL DETAILS.

S1.00, S1.00A	A1.00, 2/A1.01
LG TOC = El 37'-0"	LG TOC = El 37'-0"
Top of Steel = El 51'-3"	Top of Steel = El 52'-0"
Vault Clear Height = 14'-3"	Vault Clear Height = 15'-0"
Vault Lid Slab = 8-1/4"	Vault Lid Slab = 7-1/2"
Vault Lid TOC = El 51'-11 1/4"	Vault Lid TOC = El 52'-7 1/2"

STORMWATER TANK CAP NOTES:

1. TOP OF CONCRETE ELEVATION 37' - 2" UNLESS NOTED OTHERWISE.

2. S8 — SLAB CONSTRUCTION SHALL BE 8" NORMAL WEIGHT CONCRETE. SEE PLAN FOR REINF.



5 ENTRY PARKING RAMP FROM STREET TO P1 PART PLAN

SCALE: 1/8" = 1'-0"

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Issues

Number	Date	Description
00	10-30-25	FACADE AND INFRASTRUCTURE PACKAGE

Revisions

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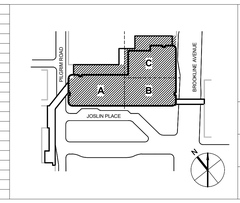
NOT FOR CONSTRUCTION



Dana-Farber Cancer Institute — Beth Israel Lahey Health / Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL

1 JOSLIN PLACE, BOSTON MA 02115

PART PLANS

Drawing Scale: 1/8" = 1'-0"

Project Number: 17000.00

Date Issued: 10/30/2025

S1.00A

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Autodesk Docs://DFCI-Cancer_Hosp/STR_Lab_DFCI_Cancer-Hospital_23-0320_R24.rvt
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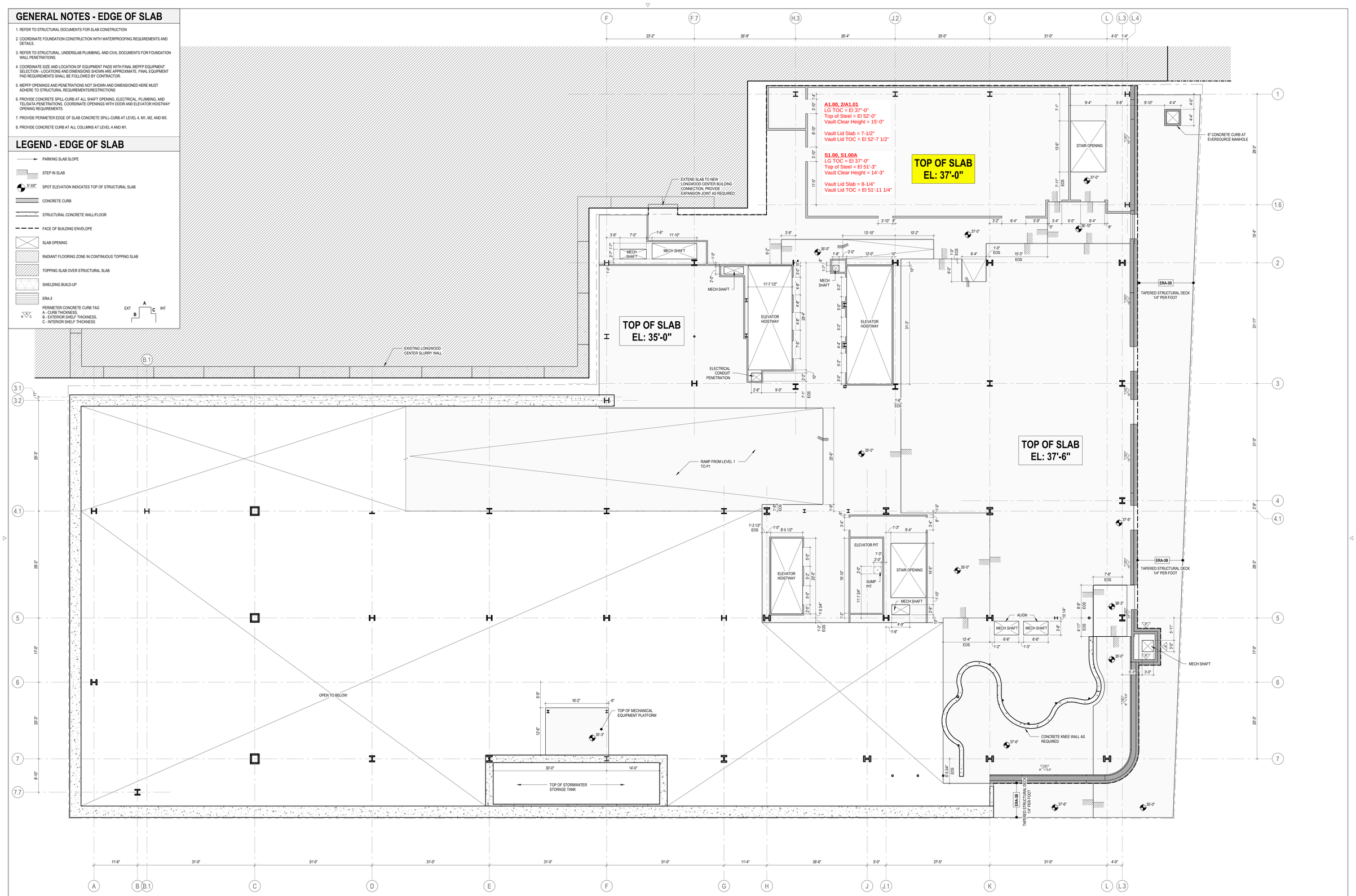
GENERAL NOTES - EDGE OF SLAB

1. REFER TO STRUCTURAL DOCUMENTS FOR SLAB CONSTRUCTION
2. COORDINATE FOUNDATION CONSTRUCTION WITH WATERPROOFING REQUIREMENTS AND DETAILS.
3. REFER TO STRUCTURAL, UNDERSLAB PLUMBING, AND CIVIL DOCUMENTS FOR FOUNDATION WALL PENETRATIONS.
4. COORDINATE SIZE AND LOCATION OF EQUIPMENT PADS WITH FINAL MEPFP EQUIPMENT SELECTION - LOCATIONS AND DIMENSIONS SHOWN ARE APPROXIMATE. FINAL EQUIPMENT PAD REQUIREMENTS SHALL BE FOLLOWED BY CONTRACTOR.
5. MEPFP OPENINGS AND PENETRATIONS NOT SHOWN AND DIMENSIONED HERE MUST ADHERE TO STRUCTURAL REQUIREMENTS/RESTRICTIONS
6. PROVIDE CONCRETE SPILL-CURB AT ALL SHAFT OPENING, ELECTRICAL, PLUMBING, AND TEL/DATA PENETRATIONS. COORDINATE OPENINGS WITH DOOR AND ELEVATOR HOISTWAY OPENING REQUIREMENTS
7. PROVIDE PERIMETER EDGE OF SLAB CONCRETE SPILL-CURB AT LEVEL 4, M1, M2, AND M3.
8. PROVIDE CONCRETE CURB AT ALL COLUMNS AT LEVEL 4 AND M1.

LEGEND - EDGE OF SLAB

- PARKING SLAB SLOPE
- STEP IN SLAB
- X'-XX" SPOT ELEVATION INDICATES TOP OF STRUCTURAL SLAB
- CONCRETE CURB
- STRUCTURAL CONCRETE WALL/FLOOR
- FACE OF BUILDING ENVELOPE
- SLAB OPENING
- RADIANT FLOORING ZONE IN CONTINUOUS TOPPING SLAB
- TOPPING SLAB OVER STRUCTURAL SLAB
- SHIELDING BUILD-UP
- ERA-3
- PERIMETER CONCRETE CURB TAG
  A - CURB THICKNESS,
  B - EXTERIOR SHELF THICKNESS,
  C - INTERIOR SHELF THICKNESS



1 GROUND - SLAB EDGE OVERALL PLAN
SCALE: 1/8" = 1'-0"

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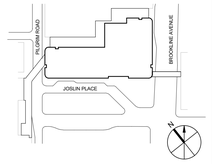

Dana-Farber Cancer Institute
Beth Israel Lahey Health
Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL
1 JOSLIN PLACE, BOSTON MA 02115

SLAB EDGE PLAN - LEVEL G

Drawing Scale: As indicated
Project Number: 17000.00
Date Issued: 10/30/2025

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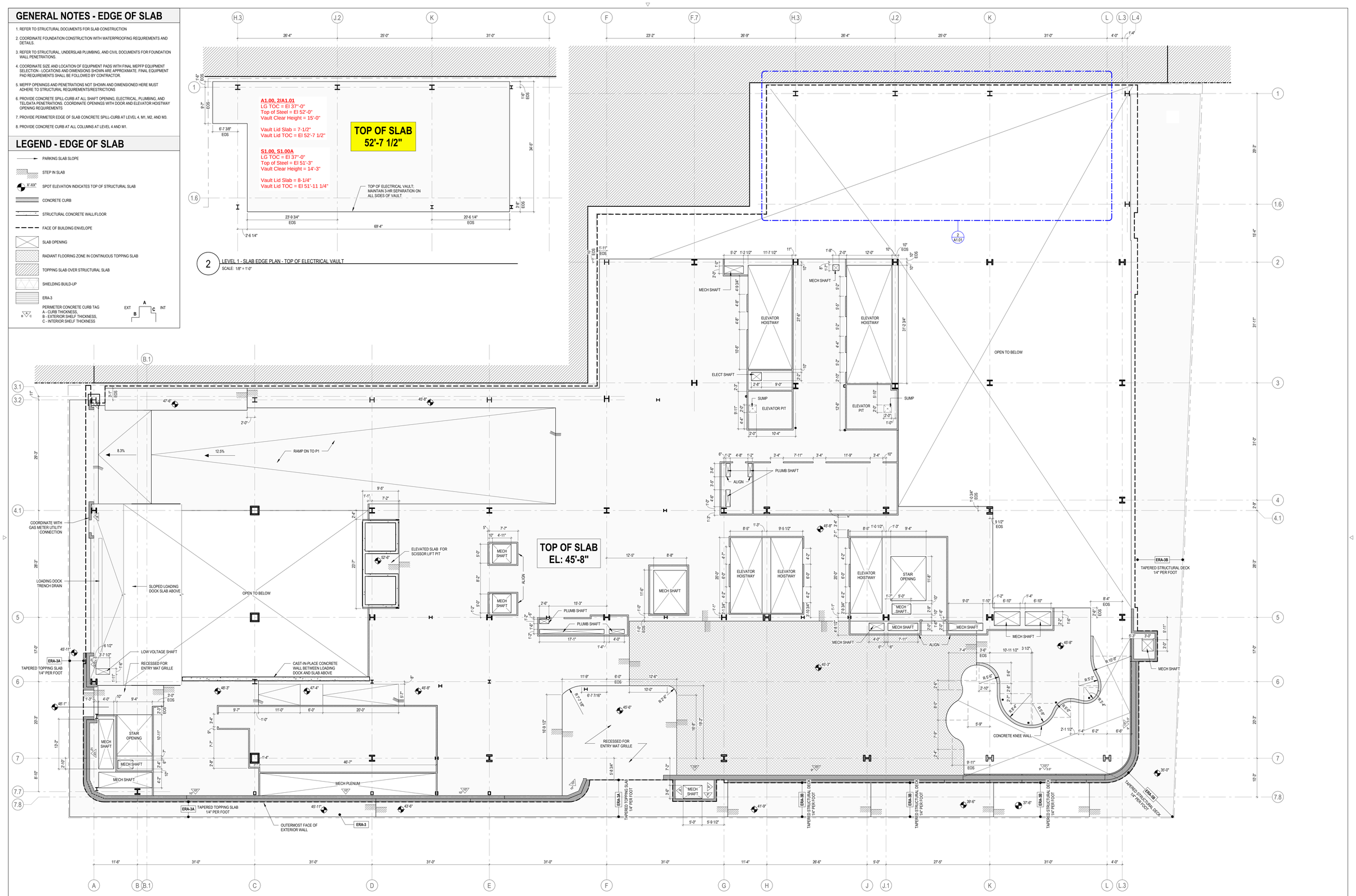
GENERAL NOTES - EDGE OF SLAB

1. REFER TO STRUCTURAL DOCUMENTS FOR SLAB CONSTRUCTION
2. COORDINATE FOUNDATION CONSTRUCTION WITH WATERPROOFING REQUIREMENTS AND DETAILS.
3. REFER TO STRUCTURAL, UNDERSLAB PLUMBING, AND CIVIL DOCUMENTS FOR FOUNDATION WALL PENETRATIONS.
4. COORDINATE SIZE AND LOCATION OF EQUIPMENT PADS WITH FINAL MEPFP EQUIPMENT SELECTION - LOCATIONS AND DIMENSIONS SHOWN ARE APPROXIMATE. FINAL EQUIPMENT PAD REQUIREMENTS SHALL BE FOLLOWED BY CONTRACTOR.
5. MEPFP OPENINGS AND PENETRATIONS NOT SHOWN AND DIMENSIONED HERE MUST ADHERE TO STRUCTURAL REQUIREMENTS/RESTRICTIONS
6. PROVIDE CONCRETE SPILL-CURB AT ALL SHAFT OPENING, ELECTRICAL, PLUMBING, AND TEL/DATA PENETRATIONS. COORDINATE OPENINGS WITH DOOR AND ELEVATOR HOISTWAY OPENING REQUIREMENTS
7. PROVIDE PERIMETER EDGE OF SLAB CONCRETE SPILL-CURB AT LEVEL 4, M1, M2, AND M3.
8. PROVIDE CONCRETE CURB AT ALL COLUMNS AT LEVEL 4 AND M1.

LEGEND - EDGE OF SLAB

- PARKING SLAB SLOPE
- STEP IN SLAB
- X'-XX" SPOT ELEVATION INDICATES TOP OF STRUCTURAL SLAB
- CONCRETE CURB
- STRUCTURAL CONCRETE WALL/FLOOR
- FACE OF BUILDING ENVELOPE
- SLAB OPENING
- RADIANT FLOORING ZONE IN CONTINUOUS TOPPING SLAB
- TOPPING SLAB OVER STRUCTURAL SLAB
- SHIELDING BUILD-UP
- ERA-3
- PERIMETER CONCRETE CURB TAG
  A - CURB THICKNESS,
  B - EXTERIOR SHELF THICKNESS,
  C - INTERIOR SHELF THICKNESS



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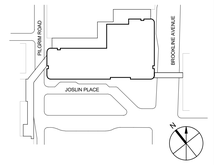

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SLAB EDGE PLAN - LEVEL 01

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EVERSOURCE	CONSTRUCTION STANDARD ELECTRIC OPERATIONS ORGANIZATION

Indoor Customer Station (STA)

Document Number:	Issued Date:	Revised Date:	Revision:	Applicability:
DTR 76.653	01-Jun-20	15-Apr-24	1	EMA

***** This Document Supersedes Eversource-EMA C10903 *****

Scope:

This document details the electrical, mechanical, and structural requirements for distribution medium voltage (below 35 kV) primary metered indoor customer stations supplied by Eversource grid.

Safety:

Providing a work environment, free of recognized hazards is a value at Eversource. Therefore, prior to the start of any work, ensure that you are familiar and knowledgeable with all Eversource Safety Rules, Policies and Procedures that are applicable to the work and tasks at hand and perform a job brief at the job site, prior to commencing work. PPE requirements to protect the worker shall be followed as required in the Eversource Employee Safety Manual.

Approved by: *Signature on File* Date: 28-Feb-24

Aymen Lpizra
Manager, T&D Standards Engineering

Rev No.	Description	Date
1	Updated in conjunction with tri-state document. See Summary of Revisions for detailed updates.	April 15, 2024
0	Issued as approved. Supersedes Eversource-EMA C10903.	June 1, 2020

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1.0 Introduction

1.1 This standard lists the minimum electrical, mechanical, and structural requirements, as well as the Customer responsibilities, for a Customer Station (STA) located within a building.

1.2 Attachment 1 in this standard lists the preferred "Material & Customer Station Accessories" for indoor Customer Station installations.

1.3 Attachment 2 in this standard includes a "Guideline for STA Customer Station Responsibilities" for Eversource and/or Customer, illustrating who furnishes, installs, maintains, and owns the aspects of the job.

1.4 Where contradictions occur from this standard and Eversource's line extension policy as established through the latest "rate-case", the line extension policy shall supersede.

1.5 For the remainder of this standard, "Customer Station" will refer to a furnished, turn-key indoor Customer Station for distribution medium voltages primary metered customers.

1.5.1 The equipment located therein is Customer owned and maintained.

2.0 Reference Documents

Eversource	DTR 76.650 Distribution Ductbank Construction & Materials
Eversource	D1200 Splicing Chamber within a Building
Eversource	SPC-HOL-006 Precast Concrete Manhole 6'0" W x 10'0" L x 8'0" H – Inside Dimensions
Eversource	SPC-SWG-005 Customer Station Metal-Clad Switchgear
Eversource	SPC-SWG-006 Customer Station Metal-Enclosed Switchgear
Eversource	SUB 010 Substation Site Development
Eversource	TD956 Station Class Installation Process for New Customer Service Solutions
ASTM	D1535-14 (2018) Standard Practice for Specifying Color by the Munsell System
IEEE	80 Safety in AC Substation Grounding (2013)
IEEE	142 Grounding of Industrial & Commercial Power Systems (1991)
M.D.P.U.	Massachusetts Department of Public Utilities No. 3A (effective March 1, 2018) Terms & Conditions of Distribution Service
NEC	National Electrical Code, Article 250, Grounding & Bonding
NESC	National Electrical Safety Code (latest version)
NFPA	National Fire Protection Association (latest version)
OSHA	Safety & Health Standards – Part 1910, Section 1920.25, Subpart D, Fixed Industrial Stairs
	Safety Color Code for Marking Physical Hazards – Part 1910, Section 1910.144, Subpart J, General Environmental Controls
State Code	Massachusetts State Building Code, 780 CMR 1007.0.

3.0 Definitions

3.1 Base Flood Elevation (BFE) – The computed elevation of flooding, including wave height, having a 1% chance (100 year) of being equaled or exceeded in any given year, as depicted on the Federal Emergency Management Agency (FEMA) Flood Insurance Rate Map (FIRM). Otherwise, flood having a 1% chance (100 year) of being equaled or exceeded in any given year.

3.2 Bonding – The connection of two or more objects together to form an electrically conductive path.

3.3 Customer – Any person, partnership, corporation, or any other entity who obtains distribution service as a customer of record of the Eversource.

3.4 Customer Station (STA) – A station class equipment request for service solution for distribution medium voltage levels (15kV, 25 kV and 35 kV) primary metered service that generally requires 100 percent Customer ownership.

3.4.1 The indoor customer station (within building) is Customer furnished, installed, maintained, and owned.

3.4.2 The *equipment* located therein is Customer furnished, installed, maintained, and owned. The Customer shall report data on equipment from inspections and tests.

3.5 Damper – A device with adjustable blades/slats motor driven by an actuator or gravity operated versions to provide a tight seal (a.k.a. shutters), used to stop or otherwise regulate airflow. Rated “fire/smoke” dampers also block out heat and smoke to contain or vent harmful fumes and slow the spread of fire.

3.6 Flood Insurance Study (FIS) – A compilation and presentation of flood risk data for specific watercourses, lakes, and coastal flood hazard areas within a community.

3.7 Foreign System – Any system or apparatus coming or introduced from the outside, and not part of, belonging to, or a characteristic of the Customer Station installation.

3.8 Grade Level (at-grade) – The reference to the grade plane representing the finished ground level adjoining the building at exterior walls (i.e., typically street access level).

3.9 Ground Mesh/Mat – A ground grid that dissipates the current flowing in from structures, or cables into the earth, and maintains surface gradients safe for human contact, i.e., step and touch potential.

3.10 Grounding (ground) – The connection of conductive parts of lines, structures, and equipment to earth or other conductive medium (grounding grid) that substitutes for earth, i.e., station ground mat conductors and rods.

3.11 Line Extension – An Underground or Overhead extension along a Public Way from Eversource’s existing electric distribution system.

3.12 Line of Demarcation – A line defining the boundary of an area of limitation, often agreed upon, to separate two distinct things.

3.13 Louver – A device with fixed, angled blades/slats that do not close, allowing light and air to pass through, but blocking out rain, direct sunlight, and noise. The device is used for air intake or exhaust.

3.14 Private Property – Travelled ways, access roads, and roads that are not defined with metes and bounds, and are not recorded as such, normally referred to as the "Customer's Property".

3.15 Public Way – Streets, roads, and ways that are defined by metes and bounds, are recorded as such in the Registry of Deeds, and are available for use by the public.

3.16 Splicing Chamber – A secure and dedicated location for splices from Eversource lines to Customer Station.

3.17 Turn-key – Relating to when the Customer Station is built, fully equipped or supplied, and completely ready for operation.

3.18 Qualified Personnel – A properly identified and authorized representative, agent, or employee of Eversource or approved Customer hired Contractor.

4.0 General

4.1 Preparation

4.1.1 Before construction begins, the Customer shall submit the Customer Station design to Eversource for review and approval.

.1.1. The Customer must receive written Eversource approval to be considered acceptable.

4.1.2 All Eversource requirements are intended to comply with all applicable municipal and state construction codes, as well as meet industry standards.

.2.1. The minimum standard requirements are those of the national, state, local municipal, and Eversource authorities.

NOTE: The Customer MUST obtain written approval from Eversource for ANY deviation from these requirements.

4.2 Design

4.2.1 Customer Stations shall be installed on the first floor of a building.

.1.1. To protect from potential flood conditions, as identified in the site's civil design requirements and flood risk data, the Customer Station may be installed on the second floor.

.1.2. See the Section 6.0, "*Customer Station Location*" for more details.

.1.3. Flood study shall be performed in accordance with **SUB 010** "Substation Site Development".

4.2.2 A Splicing Chamber is to be installed within the building as required by Distribution Engineering.

.2.1. For requirements to install a cable splicing chamber, refer to Design Standard **D1200** "*Splicing Chamber within a Building*".

.2.2. A public or private manhole may be used as an alternative if determined by Eversource.

a. For requirements to install manhole, refer to material specification **SPC-HOL-006**.

4.2.3 All six (6) sides of the Customer Station (floor, ceiling, walls) shall be approved for Class A, 3-hour fire rating as defined by the NEC.

4.2.4 The Customer shall provide construction drawings stamped by a licensed Massachusetts Professional Engineer for Eversource approval, for example:

.4.1. Site Drawings

.4.2. Conduit Routing/Connection Drawings

.4.3. Customer Station Layout Drawings (i.e., dimensions, clearances, physical location)

.4.4. Customer Station and Electrical Equipment Drawings (i.e., plan, front, & side views)

.4.5. Customer Station Grounding System Drawings

.4.6. Building Grounding Riser Diagram

.4.7. One-line Drawing

.4.8. Three-line Drawing

.4.9. Elementary Schematics

.4.10. Tripping Logic

.4.11. Automatic Bus Restoral (ABR) Scheme

.4.12. Coordination Study

4.3 Foreign systems

4.3.1 Any foreign system, not part of the installation, shall NOT enter or pass through the Customer Station (i.e., pipe or duct systems, armored or unarmored cables).

4.3.2 No materials shall be stored in the Customer Station.

4.3.3 No permanent or temporary obstructions shall be in doorways.

4.4 Security

4.4.1 It is the responsibility of the qualified personnel to ensure that the Customer Station is always locked and secured.

4.4.2 The Customer Station shall be accessible ONLY to qualified personnel.

5.0 Responsibility

5.1 Line of Demarcation

5.1.1 Eversource owns and maintains all cables and equipment up to and including the primary metering equipment, which shall be used as the line of demarcation unless a private manhole utilized.

5.1.2 Any deviation for the line of demarcation shall be approved by Eversource.

5.2 Eversource shall:

5.2.1 Design Underground or Overhead line extension needed to interconnect the Customer’s property from the Public Way.

5.2.2 Furnish, install, maintain, and own the primary cable, cable pull, cable splices, and conduit run from the manhole or splicing chamber.

.2.1. The Customer maintains and owns the splicing chamber. See Attachment 2 for responsibilities.

5.2.3 Approve the manhole or splicing chamber location and installation, as applicable.

5.2.4 Provide switchgear equipment specifications for the Customer Station location.

NOTE: The Customer Station (switchgear) must be approved by Eversource.

5.2.5 Notify the Customer and perform all repairs to Customer owned cables within the building, but at the Customer expense.

5.3 The Customer shall furnish, install, maintain, and own the following:

5.3.1 Customer Station structure including the floor, ceiling, walls, sills/risers, curbs, all ventilation openings, equipment access, and other installation fixtures and accessories.

5.3.2 Switchgear equipment in the Customer Station.

.2.1. The Customer will install their own equipment.

.2.2. The Customer will inspect and test their own equipment.

.2.3. The Customer will repair their own equipment.

5.3.3 Doorways, doors, door hardware, doorframes and hinges, panic hardware, and the lock set (except lock cylinders).

.3.1. Lock cylinders for doorways will be furnished and installed by Eversource, but at the Customer's expense.

5.3.4 Ventilation ducts, flexible fan connections, dampers, louvers, and screens, as required.

5.3.5 Lighting and electrical facilities in the Customer Station including distribution bus, fuses, insulators, connectors, load break devices, lighting fixtures, switches, power receptacles, metering conduit, and wiring in electrical conduit.

5.3.6 Sump pump, when required by the Customer Station installation design.

5.3.7 Stairs, as deemed necessary by Eversource, shall be in accordance with the most restrictive and applicable portions of both codes below:

.7.1. OSHA Safety & Health Standards – Part 1910, Section 1920.25, Subpart D, "Fixed Industrial Stairs".

.7.2. Massachusetts State Building Code, 780 CMR 1007.0.

5.3.8 Grounding system entirely, including bonding, ground mesh, ground rods, and ground bus.

5.3.9 Demarcation cable location "site" within the building at grade level, known as the splicing chamber or manhole.

.9.1. Eversource shall make splices inside the chamber at the cost of the Customer. See Attachment 2 for responsibilities.

5.3.10 Primary (high voltage) cable conduit and cable pulling from the Splicing Chamber to the Customer Station room and all necessary supports are to be done by the Customer.

5.3.11 Secondary (low voltage) cable conduit, cable pulling, and approved lugs from the Customer Electric Room to the transformers inside the Customer Station and all necessary supports.

5.4 The Customer shall:

5.4.1 Work with Eversource to obtain all permits, inspections, reports, easements, and other necessary approvals and submit them in writing to Eversource.

5.4.2 Conform to the requirements of Eversource's Standards for interconnection and/or such further requirements as Eversource may disseminate as appropriate and as approved by the M.D.P.U.

5.4.3 Responsible to install the non-electrical facilities portion of the system and to perform the necessary excavation and backfilling in accordance with Eversource's Standards for Underground Construction.

.3.1. All installations to be inspected by Eversource.

5.4.4 Coordinate with Eversource for the location, number, and type of cables and conduits.

5.4.5 Discuss and confirm with Eversource on the requirements regarding the primary cable utilized, support, pulling, termination, demarcation of ownership, and installation.

5.4.6 Furnish and install an "additional" Eversource manhole on their property or within the easement at the discretion of Eversource.

NOTE: 1. The Customer will furnish and install, at their cost, the required additional manhole(s). See Attachment 2 for responsibilities.

2. Splices will be made by Eversource in an additional Eversource manhole, at the expense of the Customer.

3. Upon completion of the project and assuming transfer of ownership, Eversource will maintain and own the required additional manhole(s).

5.4.7 Determine their contributions and communicate them to Eversource via the Eversource Customer Account Executive (CAE).

.7.1. Additional contributions, besides the required deposits, may be required.

.7.2. All Customer contributions shall be collected prior to energization.

5.4.8 Before energization, provide any certified test results that are required by Eversource.

6.0 Indoor Customer Station Location

6.1 The Customer Station shall be located inside the building and shall be approved by Eversource.

6.2 The Customer Station shall be designed to be, at a minimum, at an elevated height such that station operation and maintenance can be readily achieved in flood conditions, as identified in the Eversource Design Standard **SUB 010** for site development.

6.2.1 The Customer shall provide a site grading plan and shall be approved by Eversource.

6.2.2 The low point of the finish grade (floor) inside the Customer Station must be at least as high as the design flood elevation, or base flood elevation (BFE).

NOTE: Any deviances from these requirements shall be approved in writing by an Eversource Director.

6.3 All Customer Stations MUST have local Fire Department approval and meet ALL Fire Department suppression guidelines, before Eversource will initiate the design.

6.4 A Splicing Chamber is to be installed at or below grade level within the building.

NOTE: For details to install a cable splicing chamber within the building entrance, refer to Design Standard **D1200** "*Splicing Chamber within a Building*".

6.5 A minimum of three (3) unimpeded means of egress shall be provided.

6.5.1 One (1) egress shall exit directly to an outside area of the building.

.1.1. An indoor dedicated staircase shall be required, when necessary (i.e., an above-grade Customer Station).

6.6 The Customer Station location must permit ventilation to the outside air.

6.7 For buildings with multiple Customer Stations, the Customer Station locations shall be separate and distributed within the building to prevent a single event in one Customer Station from impacting the others.

7.0 Customer Station Size & Clearances

7.1 The Customer Station requirements, including equipment and size requirements, shall be approved by Eversource.

7.1.1 First, the equipment requirements shall be agreed to by both the Customer and Eversource.

7.1.2 Second, the size requirements shall then be agreed to by both the Customer and Eversource.

7.2 The Customer Station ceiling height shall be a minimum of 15-feet.

7.2.1 Nothing shall hang from the ceiling below 8 feet.

7.3 Equipment of any kind shall NOT be installed above the switchgear.

7.4 Switchgear clearances shall be as follows:

7.4.1 Above: A minimum distance of 4-feet shall be maintained from the top of the equipment to the ceiling of the Customer Station.

7.4.2 Front & Back: A minimum distance of 10-feet shall be maintained.

7.4.3 Two sides (Left/Right): A minimum distance of 5-feet shall be maintained on both sides of the equipment to the walls of the Customer Station.

7.5 The Customer Station shall be sized to permit the equipment to be installed and removed, so that any part of the equipment can be replaced without disturbing the remaining equipment.

8.0 Accessibility

8.1 The Customer shall provide unimpeded, 24-hours a day, outside access to the Customer Station for all authorized Eversource personnel, during the construction and after the installation is completed.

8.1.1 Eversource requires immediate and unscheduled access to the Customer Station based on system conditions.

8.2 The Customer shall provide a means of removal for the largest piece of equipment from the Customer Station location upon agreement with Eversource.

8.2.1 Means of removal shall be an exterior roll-up door.

8.2.2 This roll-up door shall be always accessible.

8.2.3 Removable panels shall not be installed on exterior façade that would impede access.

8.3 After the electrical equipment within the Customer Station has been energized, access to the Customer Station shall ONLY be by qualified personnel.

8.3.1 It is the responsibility of the qualified personnel to ensure that the Customer Station is always locked and secured.

8.4 In the event the Customer Station is de-energized, the emergency generator shall allow access to Eversource for repairs or switching operations.

9.0 Doors

9.1 The Customer Station is required to have three (3) unimpeded means of egress.

9.1.1 One (1) door shall be locked and keyed (M-key type) with appropriate hardware for Eversource use only.

9.2 All door types shall be approved for Class A, 3-hour fire rating as defined by the NFPA.

9.3 Door types & requirements:

9.3.1 Personnel doors shall be:

.1.1. Approved by the Underwriters Laboratories (UL), along with its doorframe, and carry such labels.

.1.2. Sized at 36-inches wide by 80-inches high.

.1.3. Hinged to open outward and away from the Customer Station.

a. All single doors shall be hung on three (3) hinges, or leaf if a double door.

.1.4. Equipped with bottom sweeps.

.1.5. Equipped with panic hardware.

.1.6. Close-fit in the doorframe and NOT overhang the door's riser or sill.

.1.7. Installed with two (2) exit signs per door in locations below.

a. One centered above the door.

b. Second located at the bottom, non-hinged side of the door.

.1.8. Installed with one (1) fire extinguisher per door with appropriate signage.

.1.9. Installed with doorway risers of reinforced poured concrete 4 inches in height.

.1.10. Have doorway thresholds installed on top of and encompass a doorway riser, so as not to create a step or tripping hazard (ramp on the vault interior).

9.3.2 Exterior roll-up door shall:

.2.1. Be sized at a minimum of 12 feet wide by 12 feet high.

.2.2. Be operated both electrically and manually.

.2.3. Have a watertight 4-inch-high concrete curb or "break-away block" installed inside of the roll-up door.

a. The "break-away block" shall be installed AFTER equipment is delivered.

NOTE: Additional sizing may be required based on equipment and maintenance requirements.

9.3.3 Ceiling hatch (entrance/exit) shall include:

.3.1. Hold bars and "grab" handlebars as an integral part of the hatch.

.3.2. Approval from Eversource, prior to the installation.

9.4 Door hardware & requirements:

9.4.1 Panic latches:

.1.1. Panic exit devices shall be a mortise lock type, heavy-duty push bar mechanism.

.1.2. The push bar shall extend the full width of the door.

.1.3. The finish shall be satin aluminum.

.1.4. The exit device shall be from the manufacturer, Adams Rite Series, or an approved equivalent.

.1.5. The exit device shall be furnished on all single and double doors.

9.4.2 Butt hinges:

.2.1. The butt hinge shall be from the manufacturer, Hager, or an approved equivalent, which is a full-service ball bearing, 4½ inches long with non-removable pins.

9.4.3 Door holders:

.3.1. The door holder shall be an overhead type with a hold-open feature from the manufacturer, Glyn-Johnson, or an approved equivalent.

.3.2. A quantity of two (2) door holders shall be installed on double doors.

9.4.4 Locks:

.4.1. The mortise lock type cylinders shall be from the manufacturer, Best, or an approved equivalent, which will be furnished by Eversource for all doors at the Customer expense.

.4.2. All door locking hardware shall be the permanently locked type.

9.4.5 Door louvers:

.5.1. The use of louvers on doors is NOT permitted, unless required by Eversource.

9.5 Doorframes shall be bonded to the Customer Station ground bus bar.

9.5.1 Bonding wire shall be green jacketed, copper stranded, #6 AWG cable.

9.6 Doorways shall NOT have any permanent or temporary barriers.

9.6.1 All doorways and/or passageways shall have NO obstructions, always.

10.0 Conduits

10.1 Conduits shall have a minimum inside diameter of 5 inches, unless otherwise specified by Eversource.

10.1.1 Each primary cable supplying the switchgear shall require two conduits. One for the installed cable and one spare.

10.2 All conduits run indoors or outdoors shall be encased in concrete.

10.2.1 Spacers shall be used to maintain correct spacing and alignment of the ducts (standard base spacers are 3 inches).

10.2.2 Outdoor conduits run through earth shall be encased in concrete with standard spacing and alignment and buried (below-grade) at a minimum depth of 36 inches.

10.2.3 See Construction Standard **DTR 76.650** "*Distribution Ductbank Construction and Materials*" for more information.

10.3 The location, number, and type of conduits shall be specified by Eversource, when the service type and requirements are determined.

10.4 The total number of bends shall NOT exceed 270-degrees.

10.4.1 The trench shall be as straight as possible from the point of termination on the building to our facilities.

10.4.2 Where the total bend radius needs to be restricted and changed due to complications in cable pulling, Eversource may determine to reduce the total bend radius.

10.4.3 Examples of conduit runs are as follows:

.3.1. Manhole to manhole.

.3.2. Manhole to splicing chamber.

.3.3. Manhole to switchgear.

.3.4. Splicing chamber to switchgear.

10.5 Any single 90-degree bend shall NOT be made of a radius less than 48-inches.

10.5.1 These sweeps shall be of the long-radius type.

10.6 All conduits shall extend 3-inches past the base of the concrete equipment pad or floor.

10.6.1 Bell ends shall be installed on all PVC conduits.

10.6.2 Ground bushings shall be installed on all metal rigid conduits.

10.6.3 Concrete shall be placed to surround and brace the extended 3-inches of conduit to provide strength for cable pulling.

.3.1. Typically, a slope of 45-degrees from the equipment pads to the top of each conduit bell end.

10.7 All conduits shall be brushed with mandrels and pulling line installed.

10.8 Conduits shall be sealed using duct sealants, duct plugs, or an approved equivalent.

10.9 When the conduits enter the Customer Station under the floor, they shall be turned up at a designated location with a 48-inch, rigid, long-radius, 90-degree bend.

10.10 There shall be a pulling iron or similar device installed opposite each bank of conduits.

10.11 Switchgear Conduit

10.11.1 From the switchgear to the remote operator, there shall be two (2), 2 inch conduit installed.

10.11.2 From the switchgear to the ground bus bar, there shall be two (2), 2 inch conduits installed.

10.11.3 The switchgear conduits shall be bottom entry encased in concrete.

10.12 Metering Conduit

10.12.1 From the switchgear to the metering panel, there shall be a quantity of two (2), 2 inch conduit installed.

11.0 Cables

11.1 Before routing cables into the Customer Station, the Customer shall furnish and install a Splicing Chamber per **D1200** or manhole per **SPC-HOL-006**.

NOTE: Eversource shall maintain and own all equipment up to the line of demarcation.

11.2 All primary and secondary cables or bus shall be installed in accordance with the latest National and Eversource standards.

11.3 Eversource and the Customer shall discuss, and agree on, issues regarding the purchase, support, splicing, termination, pulling, and installation of cables.

11.4 Primary cables:

11.4.1 All primary cables shall be copper.

11.4.2 All primary cable jackets shall be low smoke, zero-halogen.

11.4.3 No polyvinyl chloride (PVC) cable jackets shall be used.

12.0 Floor

12.1 The Customer Station floor shall be approved for Class A, 3-hour fire rating as defined by the NFPA.

12.2 The material of the floor shall be of reinforced concrete with adequate thickness for overall loads imposed by equipment and any rigging that may be required during installation, in-service, and removal.

12.2.1 The Customer shall design the floor to meet all current building codes and fire ratings.

NOTE: A floor shall NOT be composed of drywall, wood, or any combustible material.

12.3 The Customer Station floor shall have the capability to contain and collect equipment fluid (i.e., transformer oil) in the event of a spill per NEC Article 450.

12.4 All doorway risers or “break-away blocks”, shall be coated and sealed with electrically safe, toxic-free, epoxy paint.

12.4.1 The paint shall be yellow in accordance with ANSI and OSHA standard.

12.5 The Customer Station floor shall be coated and sealed with slip-resistant, electrically safe, toxic-free, epoxy paint.

12.6 Under certain conditions and/or in some locations, such as a Customer Station that is below-grade, it may be necessary to provide a sump and sump pump.

12.6.1 If a sump pump is required, it shall be provided with an AC Volt outlet at the pump and equipped with an audible “High-liquid Level” alarm inside the Customer Station.

.1.1. The Customer Station floor shall be pitched ¼ inch per foot toward the floor’s sump pump or drain, if applicable.

.1.2. The pump shall be connected to the emergency power with battery backup.

13.0 Equipment Pads

13.1 The switchgear shall have a 4 inch high concrete pad, referred to as an “equipment pad” in this standard or also a “housekeeping pad” in some drawings.

13.1.1 Equipment pads shall extend 4 inches past the equipment footprint on all sides.

13.1.2 All exposed corners and edges shall have a minimum of ¾ inch chamfer.

14.0 Ceiling & Walls

14.1 All ceiling and wall materials shall be approved for Class A, 3-hour fire rating as defined by the NFPA.

14.1.1 Ceiling metal decking is required to be 3-hour fire-proofed and sealed.

14.1.2 Wall structural steel members, when framed into the Customer Station walls, shall have its ends protected against fire.

14.2 The ceiling and wall materials shall be of reinforced concrete with adequate thickness for overall loads imposed by equipment and any rigging which may be required during installation, in-service, and removal.

14.2.1 The Customer will design the ceiling and walls to meet all current building codes and fire ratings.

NOTE: The ceiling, walls, or structural support members shall NOT be composed of wood, drywall, or any combustible material.

14.3 Supports for Customer Stations shall be:

14.3.1 Foundation of masonry or reinforced concrete.

14.3.2 Construction of masonry, reinforced concrete, or structural steel.

NOTE: Wood shall NOT be used as the material in the structural support members for the ceiling or walls.

14.4 All walls shall be coated and sealed with electrically safe, toxic-free, epoxy paint.

14.4.1 The paint shall be white in accordance with ANSI standard.

14.5 All ceilings made of concrete shall be coated and sealed with electrically safe, toxic-free, epoxy paint.

14.5.1 The paint shall be white in accordance with ANSI standard.

15.0 Ventilation

15.1 System Type

15.1.1 The ventilating system shall be:

.1.1. Ventilated by a forced air circulation system and include, but not limited to, proper ducts, openings, covers, and fans.

.1.2. Capable to meet the following Customer Station conditions:

a. A maximum ambient temperature, not exceeding 104°F.

b. An average temperature for any 24-hour period, not exceeding 86°F.

.1.3. Based on the secondary main switchgear size.

15.1.2 A 3-position control for the ventilation system shall be provided with manual, auto, and timer positions.

.2.1. The control shall be installed on the wall at an accessible height.

15.1.3 When operating properly, the ventilation system shall be adjusted to NOT cause a vacuum such that the doors cannot be opened manually.

15.2 Ducts

15.2.1 The ventilating ducts shall be approved for Class A, 3-hour fire rating as defined by the NFPA.

15.2.2 Where ducts connect to a power drive fan housing, the connection joint shall be of the flexible type.

15.2.3 Ducts shall not be lower than 8 feet from the ceiling.

15.3 Openings

15.3.1 Location of ventilation openings shall be as far away as possible from doors, fire escapes, and combustible materials.

.1.1. The smallest size openings shall be utilized.

.1.2. The openings shall be arranged with diagonal corners for maximum airflow across all equipment in the Customer Station.

.1.3. The supply opening shall be mounted low.

.1.4. The exhaust outlet shall be mounted high.

15.3.2 Inside the Customer Station: All ventilation openings shall be covered with an adjustable, motor driven damper.

.2.1. The material of the damper shall be aluminum or steel.

.2.2. The size of the damper shall be selected to ensure full airflow without an increase in air velocity. See Section 15.4, "*Fan & Motor*", for details.

.2.3. The damper shall be equipped with a screen and an actuating device to shut the blades for a tight seal.

a. The actuating device will function to automatically close the intake, when the fire system (i.e., heat and smoke detector) is activated.

b. The design shall be submitted to Eversource for approval.

15.3.3 Outside the Customer Station: All ventilation openings shall be covered on the outer side by a louver.

.3.1. The material of the louver shall be aluminum, copper, or galvanized steel.

.3.2. The size of the louver shall be selected to ensure full airflow without an increase in air velocity. See Section 15.4, "*Fan & Motor*", for details.

.3.3. Fixed blades on the louver shall be set at 45 degrees and backed with a removable screen.

15.3.4 Screen material, used with either a damper or louver, shall be made of aluminum, copper, or galvanized steel.

.4.1. The wire screening size shall be a minimum of 1/8 inch to a maximum of ¼ inch.

15.4 Fan & Motor

15.4.1 The ventilating fan capacity shall be selected for the maximum capacity of service equipment in the Customer Station.

.1.1. If the initial installation was at a lower maximum capacity, then the ventilation system shall be updated.

15.4.2 Air velocity of the ventilation system shall not exceed 3,000 feet-per-minute. The design should not exceed 2,000 feet-per-minute.

15.4.3 Discharge pressure of any ventilating fan shall be a minimum of ¾ inches of water.

15.4.4 The motors furnished with each ventilating fan shall be designed for the available voltage.

15.4.5 Location of each ventilating fan and its motor will depend on space limitations and shall be approved by Eversource.

.5.1. The controls for each motor shall always be located inside the Customer Station.

NOTE: The motor shall NOT be installed over the switchgear equipment, if applicable.

15.4.6 Rigid supports shall be used to securely mount or fasten both the fan and its motor.

.6.1. If no rigid supports are in the ventilation system design, a concrete pedestal of ample size (to mount both fan & motor) shall be furnished by the Customer.

a. The concrete pedestal shall be equipped with anchor bolts or cinch anchors for attaching the fan and its motor.

15.4.7 The control for each motor shall consist of a magnetic starter with both manual and timer, adjustable thermostatic controls.

15.4.8 All motors are to be interconnected to the emergency generator (if applicable) with an automatic throw-over system.

15.4.9 Exhaust motor shall operate continuously upon activation of the Customer Station fire system (i.e., heat & smoke detector), exhaust dampers must remain open.

.9.1. Intake exhaust motor (if applicable) must turn off upon activation of the fire system, along with automatically closing intake dampers.

.9.2. The Customer Station heat and smoke detectors shall activate an alarm on an annunciator panel, whose location will be approved by Eversource.

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15.4.10 For determining the ventilating fan size, the following Table 1 data shall be used:

Table 1 – Service Heat Loss Data

Service Size (amps)	Service Voltage (volts)	Power (watts)
800	120 / 208	3,000
800	277 / 480	4,500
1200	120 / 208	4,500
1200	277 / 480	7,500
1600	120 / 208	4,500
1600	277 / 480	8,300
2000	120 / 208	7,500
2000	277 / 480	8,300
2500	120 / 208	7,500
2500	277 / 480	17,400
3000	120 / 208	8,300
3000	277 / 480	17,400
4000	120 / 208	15,000
4000	277 / 480	18,800
5000	120 / 208	15,000
5000	277 / 480	36,600
6000	120 / 208	17,400
6000	277 / 480	47,500

16.0 Metering

16.1 A remote meter panel shall be installed according to local Eversource Metering Standards and the requirements herein.

16.1.1 The location of the remote meter panel & cabinet shall be approved by Eversource Meter Engineering.

16.1.2 A sketch of the meter panel shall be provided to the Customer by Eversource Meter Engineering

16.2 Metering Conduit

16.2.1 From the switchgear to the metering panel, there shall be a quantity of one (1), up to 2 inch conduit installed.

16.3 The Customer shall provide one (1), up to 2-inch sized rigid conduit for wiring from the terminal block location in the switchgear metering cubicle to the remote meter panel for the extension of the secondary wiring.

16.3.1 This conduit requirement shall NOT include pull boxes, splice boxes, or similar equipment.

16.4 The Customer is to supply and install a dedicated communication connection at Eversource's discretion at the meter panel board.

16.5 The Customer is to install a dedicated 20 Amp, 120 VAC circuit located at the meter panel board.

16.5.1 The circuit breaker for this circuit shall be equipped with a circuit breaker lock.

16.5.2 See more about power receptacles in the Section 19.6, "*Electrical Facilities*".

16.6 Eversource shall supply additional metering details for equipment during design meetings to the Customer.

17.0 Customer Station Grounding

17.1 General

17.1.1 The Customer shall supply the Customer Station grounding system design stamped by a licensed Massachusetts Professional Engineer.

.1.1. The grounding system shall be based on IEEE 80, and the latest applicable industry standards.

17.1.2 The Customer shall bond to ground any piece of equipment or unit whose height is less than 10 feet from the Customer Station floor.

.2.1. At any installed height, the Customer shall bond all HVAC equipment or ducts, electric heaters, along with doorframes and the operator's workspace.

17.1.3 All ground connections shall be exothermically bonded and irreversible. Compression bonding shall NOT be used (no set-screw type).

.3.1. Bonding by permanent BurndyWeld connections is allowed.

.3.2. Bonding by permanent Swage System Technology is allowed.

17.1.4 All connections between the ground mesh, ground rods, and ground bus bar shall be a copper-to-copper path. Building steel is NOT permitted.

17.1.5 At least one (1) 500 kcmil cable shall be connected from the Customer Station to the building steel.

17.1.6 At least one (1) 500 kcmil cable shall be connected from the Customer Station to the building's ground system.

17.1.7 Resistance of the grounding system shall comply with the NEC and IEEE for grounding in Customer Stations.

.7.1. The resistance of the grounding system to earth shall be less than 25 ohms.

17.2 Customer Station Grounding at Grade Level

17.2.1 This section is for grounding when a Customer Station room is installed at grade level.

.1.1. For Customer Stations installed above-grade level, the requirement is to have installed beneath them a "cable ring" as described in Section 17.3, "*Customer Station Grounding at Above-grade Level*".

NOTE: Reminder that Customer Stations shall be designed and located to the BFE, unless otherwise approved by a Eversource Director at a minimum.

17.2.2 A grade level Customer Station grounding system SHALL have ground rods installed inside the Customer Station and bonded to its ground bus bar.

.2.1. The ends of the ground rods shall extend 6-inches above the floor to allow attachment to the ground bus bar.

.2.2. All connections shall be exothermically bonded.

17.2.3 A ground mesh/mat shall be installed.

17.2.4 For ground rod requirements see Section 17.5.

17.3 Customer Station Grounding at Above-grade Level

17.3.1 An above-grade level Customer Station grounding system shall NOT have ground rods installed inside the Customer Station.

.1.1. The ground rods shall be outside the Customer Station, down the applicable number of floors to be installed at the lowest excavated sub-grade and located effectively below the Customer Station room.

a. Above-grade level Customer Stations installed on any floor shall have the ground rods installed at the lowest excavated sub-grade.

.1.2. A “cable ring” shall be made with the same dimensions (footprint) of the Customer Station room above.

.1.3. Create the ring with bare 4/0 AWG cables between these ground rods at grade level with all connections exothermically bonded.

.1.4. From the 4/0 AWG cable ring, four (4) 500 kcmil cables shall be run up to the above grade Customer Station.

.1.5. At least two (2) independent paths shall be used to connect the four (4) 500 kcmil cables to the above grade Customer Station.

a. Each corner of the above grade Customer Station shall have a single 500 kcmil cable bonded to its ground bus bar.

17.3.2 A ground mesh/mat SHALL be installed.

17.4 Ground Mesh/Mats

17.4.1 The ground mesh shall be wire grounding mesh with 24 inch squares made from bare #6 AWG solid copper cables.

.1.1. The prefabricated copper ground mesh must be approved by Eversource.

.1.2. The Customer shall install the ground mesh according to the latest manufacturer specifications.

NOTE: A ground mesh is ONLY required if no grounds rods are installed inside the Customer Station (i.e., above-grade level Customer Station).

17.4.2 The Customer shall connect the ground mesh to the ground bus bar.

.2.1. The ground mesh shall be exothermically bonded to the one continuous ground bus bar.

.2.2. Where applicable, every ten (10) feet a bare 4/0 AWG copper cable shall connect the mesh to the ground bus bar.

NOTE: If fewer than two (2) cables are warranted per side, then the quantity must be approved by Eversource.

17.5 Ground Rods

17.5.1 The ground rods shall be copper-clad type with a minimum single length of 10 feet and a diameter of ½ inch or ¾ inch.

17.5.2 A minimum quantity of 12 ground rods shall be installed in the Customer Station.

.2.1. Corners = One at each of the four (4) corners.

.2.2. Midpoints = One at each of the four (4) midpoints between the respective corners.

.2.3. Long-sides = Two additional on each long side between the midpoint and each corner (2 + 2 totaling 4).

17.5.3 Spacing between each ground rod shall NOT exceed 15 feet around the perimeter of the Customer Station.

17.5.4 Eversource shall be notified when the ground rod installation is complete.

.4.1. After notification, Eversource will take ground resistance measurements.

NOTE: Eversource SHALL perform and approve ground resistance measurements before the pouring of any concrete or backfilling operations and prior to cadwelding the 4/0 AWG ring

17.6 Ground Bus Bar

17.6.1 The ground bus bar shall be made of copper with minimum dimensions of 2 inches wide and ¼ inches thick.

17.6.2 The ground bus bar shall be run 6 inches above the floor of the perimeter of the Customer Station.

17.6.3 Mounting shall be with round standoff insulators, sleeve anchors, and button-head screws spaced per manufacturer's recommendation.

.3.1. The standoff insulators must be from the manufacturer, Mar-Bal Inc. (see Attachment 1), or an approved equivalent.

.3.2. Do NOT use PVC material for standoffs.

17.6.4 Bus bar connections shall be one of the following:

.4.1. Overlapped, tinned, bolted, and sweated or.

.4.2. Overlapped, tinned, and double-bolted.

a. Bolts to be 3/8 inch stainless-steel and/or silicone bronze.

17.6.5 The Customer Station's ground bus bar shall connect to two (2) building locations:

.5.1. From the Customer Station to the building steel with at least one (1) bare 500 kcmil cable.

.5.2. From the Customer Station to the building's ground grid with at least one (1) bare 500 kcmil cable.

17.7 Switchgear Grounding:

17.7.1 The Customer shall supply two (2) bare 500 kcmil cables to be used for switchgear grounding.

.1.1. The Customer shall exothermically bond the cables to the Customer Station ground bus bar.

.1.2. The cables shall be installed below-grade and encased in concrete.

.1.3. The Customer shall provide at least 3 feet length of cable slack that extends above the switchgear equipment pad to allow for switchgear grounding.

18.0 Customer Station Accessories

18.1 Telephone & Telephone Line

18.1.1 The Customer shall supply two (2) dedicated, voice-grade telephones with telephone lines in locations approved by Eversource.

.1.1. One telephone and line shall be mounted next to the Operator's Workspace (i.e., desk & storage cabinet), typically near a personnel door.

.1.2. The second telephone and line shall be mounted on the metering panel.

18.1.2 The telephones and telephone lines shall be operational prior to energization.

18.2 Operator's Workspace

18.2.1 A wall-mounted desk and wall-mounted storage cabinet shall be installed inside the Customer Station in a location approved by Eversource.

.1.1. The desk size shall be approximately 34 inches wide by 30 inches deep.

a. The desk must be from the manufacturer, Tennsco (see Attachment 1), or an approved equivalent.

.1.2. The storage cabinet size shall be approximately 32 inches high by 20 inches wide by 14 inches deep. It shall be pad-lockable and have two (2) adjustable shelves.

a. The cabinet must be from the manufacturer, Durham Manufacturing (see Attachment 1), or an approved equivalent.

18.2.2 Both the desk and storage cabinet shall be bonded to the Customer Station ground bus bar with a green jacketed, copper stranded, #6 AWG cable.

18.3 Fire Extinguisher

18.3.1 The Customer shall supply and install one (1) wall-mounted, 20 pounds, Class A:B:C fire extinguisher adjacent to each personnel access door, along with appropriate signage.

18.3.2 Each fire extinguisher is to be inspected and recorded to Eversource's Operating Division.

18.4 Exit Sign

18.4.1 There shall be two (2) exit signs installed for each personnel door:

.1.1. One centered above the door.

.1.2. Second at the bottom, non-hinged side of the door.

18.4.2 Exit signs shall be illuminated.

18.4.3 The exit signs shall be connected to the emergency power with battery backup.

18.5 Heating

18.5.1 The “heating system” may consist of multiple heating units.

18.5.2 The number of heating units shall match the number of transformers in the Customer Station, at a minimum.

18.5.3 The heating unit(s) shall be fixed, wall-mounted, and standalone with an electric heating element.

18.6 Switchgear Remote Operator Box

18.6.1 The switchgear remote operator box shall be wall mounted.

18.6.2 Location shall be indicated on the drawings and approved by Eversource.

18.7 Noise/Vibration Reduction Mats

18.7.1 The switchgear located in the Customer Station may be installed on top of a noise/vibration reduction mat, approved by Eversource.

19.0 Electrical Facilities

19.1 Electrical facilities shall include all Customer Station lighting fixtures, power receptacles (GFI & non-GFI), emergency lighting, and the fire system.

19.1.1 Emergency receptacle plates shall be painted the color RED.

19.2 All electrical facilities shall be interconnected with the emergency generator service.

19.3 Separate branch circuits shall be provided for both the lighting fixtures and the power receptacles.

19.4 All wiring shall be installed in metal rigid non-PVC conduit.

19.4.1 The conduit shall be bonded to the ground bus bar.

19.5 Lighting Fixtures

19.5.1 Lighting shall be maintained always inside the Customer Station, during the construction and after the installation is completed.

19.5.2 At least one (1) lighting fixture shall be installed on each wall.

.2.1. Spacing of fixtures shall not exceed a factor of 0.7 of the mounted height.

19.5.3 All fixtures shall be furnished with LED light bulb(s).

19.5.4 All fixtures shall be controlled by a proper multiple-way switch located inside the Customer Station at each entrance.

19.6 Power Receptacles

19.6.1 All power receptacles in the Customer Station shall be 20 Amp, 120 VAC, grounded duplex receptacles from an Eversource approved manufacturer.

19.6.2 GFI Type

.2.1. At least one (1) GFI receptacle, capable of ground fault protection, shall be installed on each wall.

19.6.3 Non-GFI Type

.3.1. There shall be one (1) non-GFI receptacle, which does NOT have ground fault protection, installed near the switchgear entrance bays.

.3.2. The location of the non-GFI receptacle shall be approved by Eversource.

.3.3. A wall-mounted sign shall be installed to clearly mark that this non-GFI receptacle does NOT have ground fault protection.

19.7 Emergency Lighting

19.7.1 Emergency lights shall be installed in the Customer Station facing the personnel doors.

19.7.2 A remote diagnostic and display panel and/or test switch shall be in the Customer Station near a personnel door.

19.7.3 All emergency lights shall be interconnected with the emergency generator service.

.3.1. If service is NOT available, the emergency lights shall be powered with batteries located outside the Customer Station.

19.8 Fire System

19.8.1 At least two (2) heat detector and two (2) smoke detector shall be installed in the Customer Station.

19.8.2 All heat and smoke detectors shall be interconnected with the emergency generator service.

19.8.3 Sprinkler systems are NOT recommended by Eversource.

19.8.4 If a sprinkler system is required in the fire system design, it shall meet the following requirements.

.4.1. Equipped with a double, interlocked, pre-action sprinkler system.

a. This means that after a predetermined detection sequence is completed, the valve opens to fill the sprinkler piping with water, then the sprinkler head is activated (i.e., ready to spray).

.4.2. The sprinkler system riser shall be equipped with a combination pneumatic, electrically active valves.

.4.3. A wet-pipe sprinkler system shall NOT be permitted by Eversource.

.4.4. A dry-agent sprinkler system shall NOT be permitted by Eversource.

20.0 <u>Battery System</u>

20.1 Battery system installation is required when circuit breakers are installed within the Customer Station.

20.2 The Customer shall provide a station battery system, consisting of batteries, battery charger, and related components and accessories.

20.2.1 Components to include, but not limited to:

.1.1. Battery main distribution panel with a main or service disconnect breaker

.1.2. Battery voltmeter

.1.3. Battery ammeter

.1.4. DC ground detector and indicator

.1.5. Battery monitor and DC under voltage alarm relay with indication

20.2.2 Accessories that shall be provided are:

.2.1. Battery rack

.2.2. Battery spill containment system

.2.3. Eyewash station

.2.4. Proper signage

20.3 The nominal DC voltage is 125 VDC but shall NOT be less than 48 VDC.

20.4 The battery system shall be adequately sized by the Customer and follow the latest industry standards and guidelines from Section 2.0, “Reference Documents”, for a high voltage substation.

20.5 The batteries shall be in a separate room or enclosure from the switchgear room or Customer Station.

20.5.1 Only with Eversource approval, may the batteries be located inside the Customer Station room.

.1.1. If approved to be installed inside the Customer Station, the batteries shall have a battery cage or plexiglass guard provided.

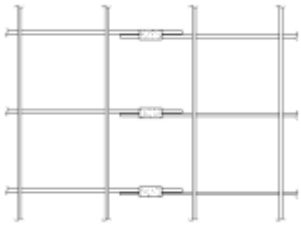
20.6 When battery ventilation is utilized, it shall NOT vent into the switchgear room or Customer Station.

21.0 Deviation

21.1 This standard sets forth the current Eversource 'best practices' for most applications of this subject matter. Therefore, deviation from this standard is not permitted. However, in unique instances a user may submit a written deviation request including justification to the listed Subject Matter Expert (SME). The SME must approve or deny the request in writing prior to the user commencing any non-standard activities. The SME may consult with his/her supervisor, co-SME if any and co-SME supervisor, and subsequently must copy any approval to them.

Attachment 1 – Material & Customer Station Accessories

Table A – Indoor Customer Station Preferred Items

Description	Manufacturer [1] p/n	Supplier p/n	Illustration
CABINET, wall-mountable with keyhole slots (no hardware included), welded 23 gauge steel, 19-7/8"W x 14-1/4"D x 32-3/4"H, includes two adjustable shelves, flush pull door, gray powder-coated finish, padlock hasp (lock not provided), cabinet weight 35 lbs, load capacity 125 lbs, shelf capacity 35 lbs.	Durham MFG 056-95	Grainger 3KR02	
DESK, wall-mountable (bracket included no hardware), 14 gauge steel, 34-1/2"W x 29"D x 17-1/2"H (add 16-1/4" to height for mounting bracket); includes sturdy writing surface, compartmentalized riser for storing writing utensils, forms and papers; drawer comes standard with lock and two keys.	Tennsco SR-59 MED GRAY	Grainger 8CFK7	
GROUND MESH, prefabricated 12 feet W x 50 feet L wire grounding mesh with 24 inch squares made from #6 AWG solid copper conductor, weight 69 lbs; includes overlapping ends with overhang two ends to allow for side-by-side connections of adjoining mats; UL listed.	Harger GM1250624OE2	TBD	
STANDOFF INSULATORS, round, aluminum, 600 volt UL rating, UNC thread style with insert size 3/8-16 x 3/8, diameter 1-3/4", height 1-3/8", thread stripping torque 30 ft. lbs, molded from a flame retardant and track resistant glass-reinforced thermoset polyester molding compound, for use with electrical bus bar.	MAR-BAL, Inc. R2137-A4	TBD	

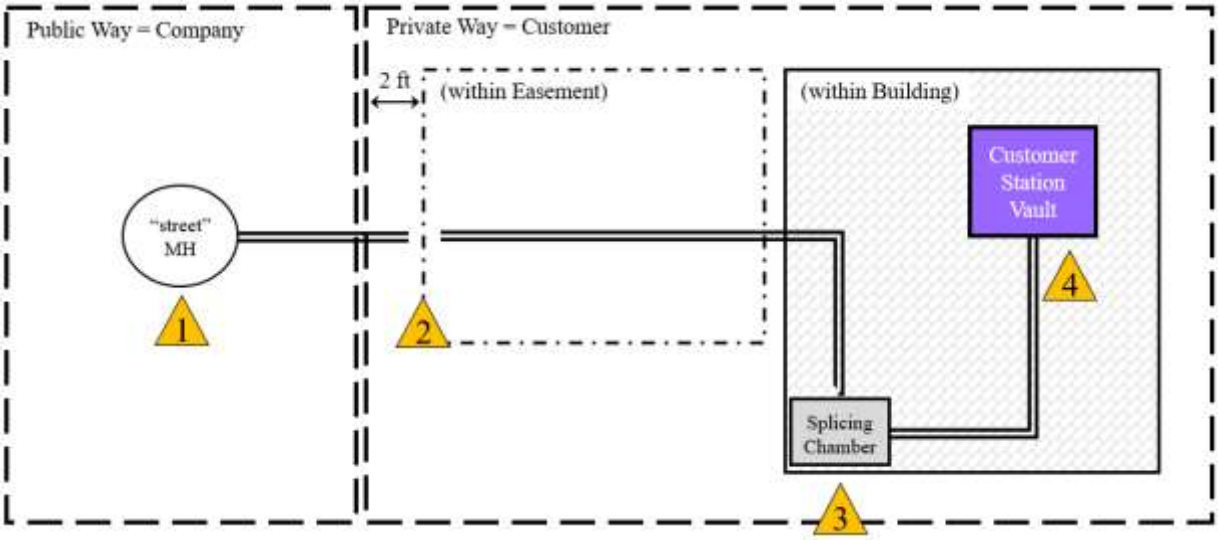
Note 1: The manufacturer part numbers listed shall be used according to this standard. Where questions or deviations arise, Eversource shall discuss and approve an equivalent alternative for the Customer.

Attachment 2 – Guideline for STA Customer Station Responsibilities

** *Where contradictions occur from this attachment and Eversource's line extension policy as established through the latest "rate-case", the line extension policy shall supersede.* **

Table AA – Guideline with No "additional" Manhole

Customer Station Vault : Primary-metered Customers : No "additional" Manhole									
#	Sequence	Work	Responsibilities NOTE 1						
			Furnish/Procure (who gets material?)	Paid for by	Install	Paid for by	Maintained	Paid for by	Owned
1	1	Splices in "Street" Manhole	ES	Customer	ES	Customer	ES	ES	ES
2	1 to 2	Conduit Run - "street" manhole to Private Way	ES	Customer	ES	Customer	ES	ES	ES
3	2 to 3	Conduit Run - NOTE 2 Private Way to Splicing Chamber	Customer	Customer	Customer Hires Approved Contractor	Customer	ES	ES	ES
4	1 to 3	Cable Pull - "street" manhole to Splicing Chamber	ES	Customer	ES	Customer	ES	ES	ES
5	3	Installation of Splicing Chamber	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
6	3	Splices in Splicing Chamber	ES	Customer	ES	Customer	ES	ES	ES
7	3 to 4	Conduit Run - Chamber to Vault	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
8	3 to 4	Cable Pull - Chamber to Vault	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
9	4	Equipment: Switchgear	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
10	4	Prepare Cables at Switchgear - (heat/cold shrink kits, stripping cable, crimping lug)	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
11	4	Terminate Cables at Switchgear - NOTE 3 (making connection, landing, bolting down)	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer



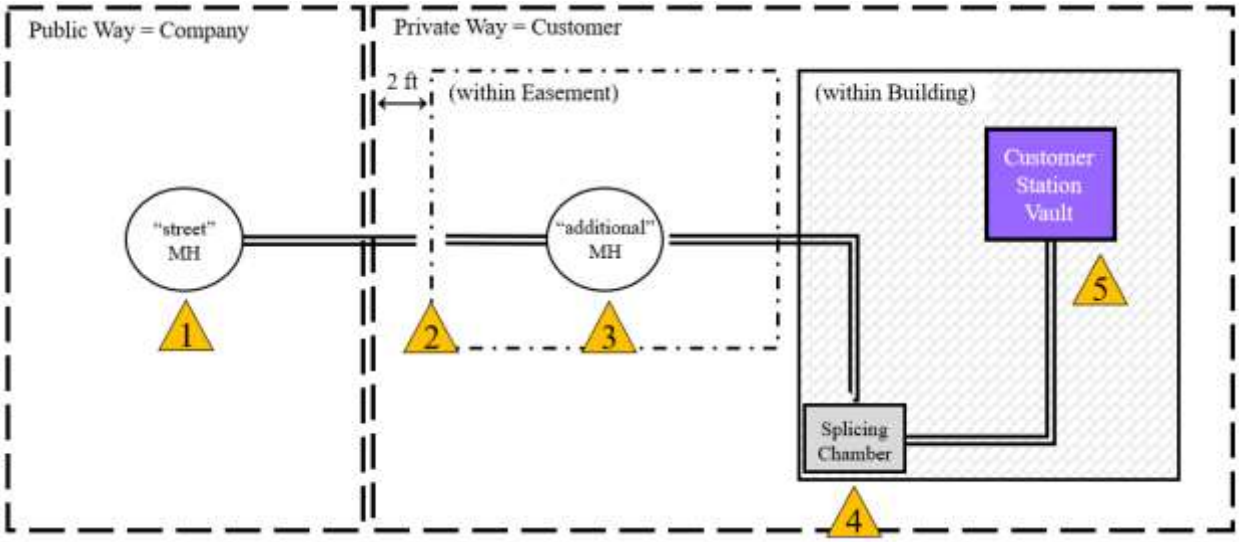
Note 1: The Customer shall discuss and concur with Eversource on issues regarding the primary cable purchase, support, pulling, termination, demarcation of ownership, and installation.

Note 2: It is assumed the Developer transferred ownership to Eversource.

Note 3: At the time of "pick-up" of station, Eversource shall verify torque at sign-off, but the Customer owns the termination.

Table BB – Guideline with “additional” Manhole

Customer Station Vault : Primary-metered Customers : with "additional" Manhole									
#	Sequence	Work	Responsibilities NOTE 1						
			Furnish/Procure (who gets material?)	Paid for by	Install	Paid for by	Maintained	Paid for by	Owned
1	1	Splices in "Street" Manhole	ES	Customer	ES	Customer	ES	ES	ES
2	1 to 2	Conduit Run - "street" manhole to Private Way	ES	Customer	ES	Customer	ES	ES	ES
3	2 to 3	Conduit Run - NOTE 2 Private Way to "additional" MH	Customer	Customer	Customer Hires Approved Contractor	Customer	ES	ES	ES
4	3	Installation of "Additional" Manhole NOTE 2	Customer	Customer	Customer Hires Approved Contractor	Customer	ES	ES	ES
5	3	Splices in "Additional" Manhole	ES	Customer	ES	Customer	ES	ES	ES
6	3 to 4	Conduit Run - NOTE 2 "additional" MH to Splicing Chamber	Customer	Customer	Customer Hires Approved Contractor	Customer	ES	ES	ES
7	1 to 4	Cable Pull - "street" manhole to Splicing Chamber	ES	Customer	ES	Customer	ES	ES	ES
8	4	Installation of Splicing Chamber	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
9	4	Splices in Splicing Chamber	ES	Customer	ES	Customer	ES	ES	ES
10	4 to 5	Conduit Run - Chamber to Vault	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
11	4 to 5	Cable Pull - Chamber to Vault	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
12	5	Equipment: Switchgear	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
13	5	Prepare Cables at Switchgear - (heat/cold shrink kits, stripping cable, crimping lug)	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer
14	5	Terminate Cables at Switchgear - NOTE 3 (making connection, landing, bolting down)	Customer	Customer	Customer Hires Approved Contractor	Customer	Customer Hires Approved Contractor	Customer	Customer



Note 1: The Customer shall discuss and concur with Eversource on issues regarding the primary cable purchase, support, pulling, termination, demarcation of ownership, and installation.

Note 2: It is assumed the Developer transferred ownership to Eversource.

Note 3: At the time of “pick-up” of station, Eversource shall verify torque at sign-off, but the Customer owns the termination.